

GREY STRUCTURE MATERIAL ESTIMATE

Double-storey house with mumty (2nd-floor room) - Plot No. 79/K

Quantities to procure (upper estimate, including wastage)

Cement 1,560 bags (50 kg)	Bricks 125,000 nos	Steel (Grade 60) 8.08 tons	Crush 4,200 cft	Sand (all) 7,750 cft
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Client	Mr. Saleem
Architect	Faisal Associates, WAPDA City, Faisalabad - working drawings sheets 01-24 (16-08-2026)
Structural engineer	Fakhar Associates - M. Fakhar Iqbal Sheikh (PEC Civil 11697), drawings 2026/L-20/S-01 to S-09 (14-08-2026)
Covered area (as per drawings)	Ground 1,898 sft + First 1,844 sft + Second 362 sft = 4,104 sft
Scope	Complete grey structure from excavation to roof parapet: foundations, plinth, brickwork, all R.C.C, steel, plaster, sub-floors, boundary walls, septic tank, manholes, overhead water tank, sewerage / rain-water / water-supply / gas pipes and electrical conduits
Estimate basis	Upper estimate - quantities measured from the vector drawing files, upper end of standard coefficient ranges, standard wastage added
Date	30 September 2026

1. Final material requirement (what to buy)

The quantities below are the total for the complete grey structure described on the cover, including standard wastage. Where a standard gives a range, the upper value has been used, as requested. "Net" is the measured quantity; "To procure" is the net quantity plus wastage, rounded up. Fill in your local rates to get the cost.

#	Material	Net qty	Waste	To procure	Unit	Remarks / purchase guide	Rate (Rs)	Amount (Rs)
1	Cement (OPC, 50 kg bag)	1,481	5%	1,560	bags	Buy fresh stock per stage; do not store more than 4-6 weeks		
2	Bricks - 1st class burnt clay (9" x 4½" x 3")	118,495	5%	125,000	nos	Standard 13.5 bricks per cft of brickwork		
3a	Steel Grade 60 deformed - 10 mm (#3)	4,406	5%	4,630	kg	Laps & hooks already included in net		
3b	Steel Grade 60 deformed - 12 mm (#4)	2,165	5%	2,280	kg	Laps & hooks already included in net		
3c	Steel Grade 60 deformed - 20 mm (#6)	976	5%	1,030	kg	Laps & hooks already included in net		
3d	Steel Grade 60 deformed - 6 mm (#2) - rings/stirrups	132	5%	140	kg	Laps & hooks already included in net		
3	Steel - TOTAL	7,679	5%	8,080	kg	= 8.08 metric tons		
4	Binding wire (annealed, 18 gauge)	77	-	85	kg	About 10 kg per ton of steel		
5	Crush / coarse aggregate (¾" down)	3,984	5%	4,200	cft	≈ 42 tractor-trolleys of 100 cft		
6	Sand for concrete (Chenab / Lawrencepur)	2,014	10%	2,250	cft	Coarse clean sand for RCC & PCC		
7	Sand for masonry mortar & plaster (Ravi)	4,415	10%	4,900	cft	Fine sand; sieve for plaster		
8	Fill sand under ground floors (4")	590	incl.	600	cft	Any clean sand / Ravi sand		
9	Earth filling (plinth + porches, compacted)	6,405	incl. 20%	6,450	cft	Re-use ~3,200 cft surplus excavated earth; import about 3,250 cft		
10	Bitumen (hot, vertical DPC 2 coats)	304	5%	320	kg	Inner face of external plinth walls		
11a	Sewerage (UPVC) 5"	156	10%	172	ft	≈ 9 lengths of 20 ft		
11b	Sewerage (UPVC) 4"	128	10%	141	ft	≈ 8 lengths of 20 ft		
11c	Sewerage (UPVC) 3"	218	10%	240	ft	≈ 12 lengths of 20 ft		
11d	Rain water (UPVC) 3"	112	10%	124	ft	≈ 7 lengths of 20 ft		
11e	Water supply (PPR) 1"-1¼"	118	10%	130	ft	≈ 7 lengths of 20 ft		
11f	Water supply (PPR) ¾"	370	10%	407	ft	≈ 21 lengths of 20 ft		
11g	Water supply (PPR) ½"	140	10%	154	ft	≈ 8 lengths of 20 ft		
11h	Hot water (PPR) ¾"	360	10%	397	ft	≈ 20 lengths of 20 ft		
11i	Hot water (PPR) ½"	80	10%	88	ft	≈ 5 lengths of 20 ft		

#	Material	Net qty	Waste	To procure	Unit	Remarks / purchase guide	Rate (Rs)	Amount (Rs)
11j	Gas (GI/PPR-gas) ½"-¾"	190	10%	209	ft	≈ 11 lengths of 20 ft		
12a	Electrical conduit (PVC) ¾"	2,900	10%	3,191	ft	≈ 320 lengths of 10 ft		
12b	Electrical conduit (PVC) 1"	1,390	10%	1,530	ft	≈ 153 lengths of 10 ft		
12c	Electrical conduit (PVC) 1½"	85	10%	94	ft	≈ 10 lengths of 10 ft		

Pipe lengths are for pipes only; fittings, manhole covers, boxes and accessories are listed in Sections 7 and 8. Optional items (roof insulation with brick tiles, underground water tank, external paving) are NOT included above - see Section 9.

2. Stage-wise requirement (order material in phases)

Buying material stage by stage reduces theft, damage and cement ageing. Quantities include wastage.

Stage	Cement (bags)	Bricks (nos)	Steel (kg)	Crush (cft)	Concrete sand (cft)	Mortar/ plaster & fill sand (cft)
1. Foundation & plinth	255	25,798	1,397	893	468	737
2. Ground-floor walls & GF roof	299	28,556	2,754	914	479	731
3. First-floor walls & FF roof	307	31,403	2,702	881	461	801
4. Mumty (2F), parapets & roof tank	149	14,974	907	344	180	405
5. Plaster, sub-floors & floor bases	358	0	0	850	445	2,029
6. External works (boundary, septic, manholes)	187	23,689	302	302	183	802
TOTAL (incl. wastage)	1,555	124,420	8,062	4,183	2,216	5,506

3. Project data read from the drawings

Item	Data used in this estimate
Structural system	Load-bearing brick masonry (9" external & main walls, 4½" partitions) with 5½" R.C.C slabs; concealed beams CB-1 (12" x 9½") and CB-2 (9" x 9½") carry first-floor walls that do not sit on ground-floor walls; one R.C.C column at the car-porch front corner
Plot / building	Plot about 37' wide x 65' deep (left boundary skewed); building 37'-3" x 51'-7" on ground floor; corner plot with main gate (front) and side gate (left)
Levels	Road ±0; car porch +1'-3"; finished floor +2'-9"; floor-to-floor 12'-0" (11'-0" clear + 1'-0" slab band); mumty 9'-0" clear; terrace parapet 2'-6"; mumty parapet 1'-6". Natural surface level "as per site" - taken 1'-6" below road as drawn
Specifications (S-01 notes)	Deformed steel Grade 60; slab & beam R.C.C 1:2:4 (3000 psi); column R.C.C 1:1.5:3 (4000 psi); covers: slab ¾", beams, columns & foundation 1½"; foundation masonry 1:6; P.C.C 1:4:8; D.P.C 1½" 1:2:4 with vertical DPC (¾" plaster 1:3 + 2 coats hot bitumen)

Item	Data used in this estimate
Foundation types (S-02)	1-1 external property-line wall (eccentric): P.C.C 2'-9" x 3", R.C.C footing 2'-6" x 6"; 2-2 external walls: P.C.C 3'-0", footing 2'-6"; 3-3 internal 9" walls: P.C.C 3'-6", footing 3'-0"; 4-4 4½" walls: P.C.C 2'-6", footing 2'-0". Footing steel #3@6"/#3@8". Brick steps 22½" (6"), 18" (6"), 13½" (12"), 9" wall; plinth beam 9"x9" 2#4+2#4, #2@10" rings; depth 2'-6" below NSL
Measured foundation lengths	Type 1-1: 40.3 ft; type 2-2: 133.4 ft; type 3-3: 86.2 ft; type 4-4: 110.8 ft; total centre-line 370.7 ft (38 wall runs - Appendix A)
Slabs (S-06/07/08)	5½" thick. GF roof 1,705.0 sft; FF roof 1,650.7 sft; mumty roof 353.2 sft (net of stair openings, open shaft and terrace)
Concealed beams	CB-1: GF roof 81.7 ft (5 beams) + FF roof 15.5 ft (1); CB-2: GF roof 48.5 ft (5) + FF roof 102.0 ft (7) + mumty roof 30.3 ft (2); 23 bed plates 9"x3"x3" (3#4)
Stair (S-03 + sheet 06/07)	Zig-zag (dog-legged), 2 flights per storey, 22 risers of 6.57", treads 10½", width 3'-6", 6" waist with #4@5" main bars, #3@9"/12" distribution, 6" landing, base footing
Lintels (S-04/S-05)	DL-1 9"x9" 4#4 #3@6"; DL-2 4½"x9" 4#4 #2@6"; WL-1 9"x9" 4#4; WL-2 9"x9" 2#4+4#4; WL-3 4½"x9"; WL&DL-1 9"x9"; VL-1 9"x9" 4#3 #3@10". Key plans show 21 (GF), 24 (FF) and 6 (2F) lintels
Wall heights used	GF 11'-0" (DPC to slab soffit); FF 11'-6.5"; mumty 9'-6.5"; terrace parapet 3'-0.5"; mumty parapet 2'-0.5" (from slab top, incl. roof build-up)
Services on drawings	Sewerage in 3"/4"/5" UPVC with 9 manholes and septic tank connected to main sewer; 4 rain-water pipes in 9"x6" ducts; cold/hot water & gas lines (sheets 21-24); electrical layout with legend (sheets 13-16); overhead water tank on mumty roof

4. How the estimate was prepared

- All 40 sheets were studied: 9 structural sheets (foundation layout & sections, stair, lintel key plans & sections, three slab reinforcement plans, beam sections) and 31 architectural/services sheets (plans, elevations, wall section, boundary walls, bath details, electrical, sewerage, sanitary).
- The structural PDFs are AutoCAD vector files, so wall lines were measured directly from the file geometry instead of scaling paper prints. Scale was calibrated on the 13'-0" x 13'-0" Bed-1 (1 ft = 33.053 drawing units); 9" and 4½" walls measured 9.0" and 4.5". The hand-built wall network (Appendix A) matches the raster-measured wall area within 1%.
- Foundation volumes use exact polygon unions of every layer (PCC, footing, each brick step, plinth), so overlaps at the many T-junctions are not double counted (unions are about 12% less than simple length x width).
- First-floor walls were measured from the lintel key plan (S-04) and openings added back; mumty walls from the 2nd-floor plan. Slab outlines, stair openings, beams and bed plates were measured on the slab sheets.
- Slab steel: every bar set on S-06/S-07/S-08 (spacing label, its distribution dimension line and bar length) was listed individually - 44 sets (GF roof), 40 sets (FF roof), 11 sets (mumty roof) - see Appendix B.
- Sewer, water and gas pipe runs were measured on the colour-coded services sheets (sheet scale calibrated separately); vertical stacks, risers and drops were added. Electrical conduits are based on a point count of the electrical plans.

4.1 Standard coefficients used (upper end of range)

Coefficient	Value	Remarks
Bricks per cft of brickwork	13.5 nos	Standard local brick with ¾" joints

Coefficient	Value	Remarks
Mortar per cft of brickwork	0.30 cft wet = 0.40 cft dry	Range 0.25-0.30 wet; upper value used
Dry volume factor - concrete	1.57	Range 1.52-1.57
Dry volume factor - mortar	1.33	
Plaster extra for joints/uneven surface	+25%	Range 20-30%
Cement bag	50 kg = 1.25 cft	
Steel weights (lb/ft)	#2 0.167, #3 0.376, #4 0.668, #6 1.502	1 lb = 0.4536 kg
Wastage	Cement 5%, sand 10%, crush 5%, bricks 5%, steel 5%, pipes 10%	Laps, hooks, cranks, chairs are inside the net steel
Mortar ratios	Foundation 1:6 (drawing); 9" walls 1:5; 4½" walls, parapets, pillars, tanks 1:4; plaster 1:4 (walls), 1:3 (ceilings, water-retaining)	Superstructure ratios not given on drawings - assumed

4.2 Assumptions (please confirm on site)

1. Natural surface level taken 1'-6" below road (drawings: "as per site"). Every 6" higher/lower NSL changes earth filling by about 820 cft and foundation brickwork slightly.
2. Foundation R.C.C footing concrete taken 1:2:4 (mix not stated).
3. Septic tank enlarged to a standard 6'-0" x 4'-0" x 5'-6" deep (internal, 2 chambers) - the drawing shows only a 4'-10" x 3'-5" symbol, small for 8 WCs.
4. Overhead water tank taken as R.C.C 1,000 gallons (8'x5'x4'-6" internal) on the mumty roof - drawings show a symbol only. If you use plastic tanks, delete Section I (saves ~34 bags cement, ~230 kg steel).
5. Back boundary wall (37'-3") and short right-side walls at the back passage and driveway are not detailed but included (9" x 6'-6" to 7'-0"). Delete if neighbours' walls already exist.
6. Boundary wall foundation (not detailed) taken as 3" P.C.C + three brick steps + 9"x6" R.C.C plinth band + 3" R.C.C coping.
7. A 5% brickwork allowance is added for front-elevation frames, fins, piers and profile-light reveals that are not shown in plan.
8. The car-porch corner (13½" x 13½") is taken as an R.C.C column (4#6, #3@6" ties) as the notes specify column concrete - costlier than a brick pillar.
9. An 8'-0" lintel is added over the 6'-6" dining opening (not shown on the lintel key plan).
10. A 3" P.C.C 1:4:8 sub-base is included over the first-floor and mumty floors to level the 4" upstand beams and cover conduits.

4.3 Not included (exclusions)

Labour; shuttering/formwork and scaffolding (contractor's items); curing water and construction electricity; all finishing works (tiles/marble, paint, woodwork, aluminium, glass, grills, railings, gates); sanitary fixtures; electrical cables, switches, breakers and DBs (only conduits and boxes included); geyser; termite treatment; and the optional items in Section 9.

5. Detailed bill of quantities with material breakdown (net, before wastage)

Each line shows the measured quantity and the materials it consumes. Section sub-totals are net; wastage is applied only once, in Sections 1 and 2.

Item	Description	Quantity	Unit	Cement (bags)	Sand (cft)	Crush (cft)	Bricks (nos)	Steel (kg)
A. Foundations & Plinth								
A1	Excavation for wall foundations (PCC footprint x 2'-6" deep below NSL), incl. 10% working space [PCC footprint 969.6 sft (types 1-1..4-4 from S-01/S-02)]	2,666.3	cft	-	-	-	-	-
A2	P.C.C 1:4:8, 3" thick under footings (widths 2'-9"/3'-0"/3'-6"/2'-6")	242.4	cft	23.4	117	234	-	-
A3	R.C.C 1:2:4 wall footing 6" thick (2'-6"/2'-6"/3'-0"/2'-0" wide) with #3@6" transverse + #3@8" longitudinal	416.8	cft	74.8	187	374	-	673
A4	Brick masonry in foundation 1:6 - stepped footing (22½", 18", 13½") + 9" wall below plinth beam	1,075.4	cft	49.0	368	-	14,518	-
A5	Brick masonry 1:6 - 9" plinth wall from plinth beam (road level) to DPC/FFL (2'-9")	744.6	cft	34.0	255	-	10,052	-
A6	R.C.C 1:2:4 plinth beam 9"x9" under all walls, 2#4 top + 2#4 bottom, #2@10" rings [plinth beam centre-line length 370.7 ft]	203.1	cft	36.4	91	182	-	569
A7	D.P.C 1½" thick 1:2:4 at FFL over plinth walls	33.8	cft	6.1	15	30	-	-
A8	Vertical DPC: ¾" plaster 1:3 + 2 coats hot bitumen on inner face of external plinth walls [bitumen approx 304 kg (0.25 kg/sft/coat)]	607.9	sft	12.6	47	-	-	-
A9	R.C.C column 1:1.5:3 at car-porch front corner 13½"x13½" (17.0 ft high), 4#6 + #3@6" ties	21.5	cft	4.9	9	18	-	77
A10	Column footing 3'x3'x1' R.C.C 1:2:4 (#4@6" both ways) + 3" P.C.C 1:4:8	9.0	cft	1.9	6	11	-	12
A11	Earth filling inside plinth (NSL to underside of sand), compacted in 6" layers - incl. 20% compaction [floor area 1363 sft x 3.54 ft; NSL assumed 1'-6" below road (drawings say "as per site")]	5,791.0	cft	-	-	-	-	-
A12	Back-filling excavated earth around foundations (re-use)	965.0	cft	-	-	-	-	-
	Sub-total A			243.2	1,095	850	24,570	1,330
B. Ground-floor sub-floor								
B1	Sand filling 4" under floors (rooms + car porches)	589.7	cft	-	590	-	-	-
B2	P.C.C 1:4:8, 3" thick sub-floor (rooms 1363 + porches 246 sft)	402.1	cft	38.8	194	388	-	-
B3	Earth filling under car porches (to porch level +1'-3")	614.2	cft	-	-	-	-	-

Item	Description	Quantity	Unit	Cement (bags)	Sand (cft)	Crush (cft)	Bricks (nos)	Steel (kg)
B4	Upper-floor sub-base: 3" P.C.C 1:4:8 over FF floor & mumty floor (levels 4" upstand CB beams, covers conduits)	436.3	cft	42.2	211	422	-	-
	Sub-total B			81.0	995	810	0	0
C. Brick masonry (superstructure)								
C-GF9	Ground floor 9" walls, 1:5 - gross 2128.5 cft (plan 193.5 sft x 11.00 ft) less openings 416.9, lintels 56.9, bed plates	1,649.6	cft	87.8	549	-	22,270	-
C-GF45	Ground floor 4½" walls, 1:4 - gross 427.3 cft (plan 38.8 sft x 11.00 ft) less openings 54.4, lintels 8.1	364.9	cft	23.3	116	-	4,926	-
C-FF9	First floor 9" walls, 1:5 - gross 2017.9 cft (plan 174.8 sft x 11.54 ft) less openings 373.4, lintels 55.2, bed plates	1,582.6	cft	84.2	526	-	21,365	-
C-FF45	First floor 4½" walls, 1:4 - gross 737.8 cft (plan 63.9 sft x 11.54 ft) less openings 91.9, lintels 13.1	632.8	cft	40.4	202	-	8,542	-
C-SF9	Second floor (mumty/room) 9" walls, 1:5 - gross 414.7 cft (plan 43.5 sft x 9.54 ft) less openings 51.0, lintels 8.9, bed plates	353.7	cft	18.8	118	-	4,775	-
C-SF45	Second floor (mumty/room) 4½" walls, 1:4 - gross 237.2 cft (plan 24.9 sft x 9.54 ft) less openings 48.4, lintels 7.2	181.6	cft	11.6	58	-	2,451	-
C-P1	Terrace parapets on FF roof (4½", one 9" run) 125.6 ft x 3.04 ft high, 1:4	154.4	cft	9.9	49	-	2,084	-
C-P2	Parapet on mumty roof 4½" 94.2 ft x 2.04 ft, 1:4	72.1	cft	4.6	23	-	974	-
C-E	Elevation features allowance 5% (front frames, fins, piers, profile-light reveals, planters) 1:4	249.6	cft	15.9	80	-	3,369	-
	Sub-total C			296.4	1,721	0	70,757	0
D. R.C.C work (1:2:4 unless noted)								
D-SGF	Roof slab 5½" - GF roof: outline 1739.1 + porch cantilever 38.7 - stair opening 72.8 (1705.0 sft)	781.5	cft	140.2	351	701	-	-
D-SFF	Roof slab 5½" - FF roof: outline 1725.0 - stair opening 74.3 (excl. open shaft & terrace) (1650.7 sft)	756.6	cft	135.8	339	679	-	-
D-SSF	Roof slab 5½" - Mumty roof: 16.85x7.75 + 17.63x12.63 (353.2 sft)	161.9	cft	29.0	73	145	-	-
D-CB-1-GF	CB-1 concealed beam 12" x 9½" in GF roof - 5 nos, 81.7 ft (4" upstand above slab; 3#6 bottom, 3#4 top, #3@7" rings)	27.2	cft	4.9	12	24	-	374

Item	Description	Quantity	Unit	Cement (bags)	Sand (cft)	Crush (cft)	Bricks (nos)	Steel (kg)
D-CB-1-FF	CB-1 concealed beam 12" x 9½" in FF roof - 1 nos, 15.5 ft (4" upstand above slab; 3#6 bottom, 3#4 top, #3@7" rings)	5.2	cft	0.9	2	5	-	71
D-CB-2-GF	CB-2 concealed beam 9" x 9½" in GF roof - 5 nos, 48.5 ft (4" upstand above slab; 2#6 top & bottom, #3@6" rings)	12.1	cft	2.2	5	11	-	225
D-CB-2-FF	CB-2 concealed beam 9" x 9½" in FF roof - 7 nos, 102.0 ft (4" upstand above slab; 2#6 top & bottom, #3@6" rings)	25.5	cft	4.6	11	23	-	443
D-CB-2-SF	CB-2 concealed beam 9" x 9½" in SF roof - 2 nos, 30.3 ft (4" upstand above slab; 2#6 top & bottom, #3@6" rings)	7.6	cft	1.4	3	7	-	131
D-LGF	Door/window/vent lintels GF - 23 nos (DL-1, DL-2, WL-1, WL-2, WL-3, VL-1 per S-05)	64.9	cft	11.7	29	58	-	264
D-LFF	Door/window/vent lintels FF - 26 nos (DL-1, DL-2, WL-1, WL-2, WL-3, VL-1 per S-05)	68.3	cft	12.3	31	61	-	287
D-LSF	Door/window/vent lintels SF - 9 nos (DL-1, DL-2, WL-1, WL-2, WL-3, VL-1 per S-05)	16.1	cft	2.9	7	14	-	73
D-BP-GF	Bed plates 9" x 3'-0" x 3" (1:2:4, 3#4) under beam ends, GF roof - 9 nos	5.1	cft	0.9	2	5	-	22
D-BP-FF	Bed plates 9" x 3'-0" x 3" (1:2:4, 3#4) under beam ends, FF roof - 12 nos	6.8	cft	1.2	3	6	-	30
D-BP-SF	Bed plates 9" x 3'-0" x 3" (1:2:4, 3#4) under beam ends, SF roof - 2 nos	1.1	cft	0.2	1	1	-	5
D-ST-GF	Zig-zag stair GF to FF: 6" waist (flights 11.65 + 9.59 ft), 6" landing 8'-3"x3'-6", 21 concrete steps; riser 6.57", tread 10½", width 3'-6"	69.2	cft	12.4	31	62	-	283
D-ST-FF	Zig-zag stair FF to 2nd floor: 6" waist (flights 11.65 + 9.59 ft), 6" landing 8'-3"x3'-6", 21 concrete steps; riser 6.57", tread 10½", width 3'-6"	69.2	cft	12.4	31	62	-	283
D-SF	Stair base footing + stem (8" slab, #4@6" both ways; stem #4@5")	12.5	cft	2.2	6	11	-	55
D-SP	P.C.C 1:4:8 under stair footing	4.4	cft	0.4	2	4	-	-
D-SH-GF	R.C.C sunshade (chajja) GF: 12.5 ft long x 1'-3" projection, 4"-3" thick, #3@6" main	5.2	cft	0.9	2	5	-	19
D-SH-FF	R.C.C sunshade (chajja) FF: 8.0 ft long x 1'-3" projection, 4"-3" thick, #3@6" main	3.3	cft	0.6	1	3	-	12
D-SH-SF	R.C.C sunshade (chajja) SF: 6.0 ft long x 1'-3" projection, 4"-3" thick, #3@6" main	2.5	cft	0.4	1	2	-	9
D-PR	Provisional: front elevation canopy / RCC fins & projections	20.0	cft	3.6	9	18	-	26
D-CP	Parapet coping 3" x 9" (1:2:4) with 2#3	41.2	cft	7.4	18	37	-	79
	Sub-total D			388.5	972	1,945	0	2,691

E. Slab reinforcement

Item	Description	Quantity	Unit	Cement (bags)	Sand (cft)	Crush (cft)	Bricks (nos)	Steel (kg)
E-GF	Slab steel GF roof (44 bar sets incl. cranked, extra top, trimmer 2#6/2#4 & chairs) - 0.84 kg/sft	1,705.0	sft	-	-	-	-	1,435
E-FF	Slab steel FF roof (40 bar sets incl. cranked, extra top, trimmer 2#6/2#4 & chairs) - 0.88 kg/sft	1,650.7	sft	-	-	-	-	1,447
E-SF	Slab steel SF roof (11 bar sets incl. cranked, extra top, trimmer 2#6/2#4 & chairs) - 0.73 kg/sft	353.2	sft	-	-	-	-	259
	Sub-total E			0.0	0	0	0	3,141
F. Plaster (grey-structure finish)								
F-IGF	Internal plaster ½" 1:4 - GF walls (gross 6044 sft, less openings, plus reveals)	5,233.7	sft	58.0	290	-	-	-
F-EGF	External plaster ¾" 1:4 - GF outer faces incl. exposed plinth	1,595.2	sft	26.5	133	-	-	-
F-IFF	Internal plaster ½" 1:4 - FF walls (gross 7262 sft, less openings, plus reveals)	6,393.1	sft	70.9	354	-	-	-
F-EFF	External plaster ¾" 1:4 - FF outer faces	1,304.0	sft	21.7	108	-	-	-
F-ISF	Internal plaster ½" 1:4 - SF walls (gross 1472 sft, less openings, plus reveals)	1,252.1	sft	13.9	69	-	-	-
F-ESF	External plaster ¾" 1:4 - SF outer faces	679.8	sft	11.3	57	-	-	-
F-P	Parapet plaster ¾" 1:4 both faces + coping	1,313.6	sft	21.8	109	-	-	-
F-C	Ceiling plaster ¾" 1:3 (slab soffits, stair soffits, shades)	3,442.2	sft	35.8	134	-	-	-
	Sub-total F			259.8	1,255	0	0	0
G. Boundary walls, gates pillars & external works								
G1	Excavation for boundary wall & pillar foundations 2'-3" wide x 2'-6" deep (141.9 ft run)	878.0	cft	-	-	-	-	-
G2	P.C.C 1:4:8 3" under boundary wall foundation	79.8	cft	7.7	39	77	-	-
G3	Brick masonry 1:6 stepped foundation (22½", 18", 13½" x 6" each)	319.3	cft	14.6	109	-	4,310	-
G4	R.C.C 1:2:4 plinth band / DPC beam 9"x6" at plinth level, 4#3 + #2@9" rings	53.2	cft	9.5	24	48	-	128
G5	Brick masonry 1:5 boundary wall 9" (122.0 ft of panels, 6'-0" to 7'-3" above road)	646.3	cft	34.4	215	-	8,726	-
G6	Brick pillars 1:4 - 7 nos 13½"x13½" + 3 nos 4'-0"x13½" feature pillars, 6'-6" high	150.9	cft	9.6	48	-	2,037	-
G7	R.C.C coping 3" on wall panels + pillar caps (1:2:4), 2#3	36.1	cft	6.5	16	32	-	44
G8	Boundary wall plaster ¾" 1:4 (both faces road side, inner face on neighbour side, pillars, top)	2,055.2	sft	34.2	171	-	-	-

Item	Description	Quantity	Unit	Cement (bags)	Sand (cft)	Crush (cft)	Bricks (nos)	Steel (kg)
G9	Lawn toe walls 9" (approx. 15 ft x 2'-6") 1:5 + 3" PCC base	28.1	cft	2.0	12	5	380	-
G10	Entrance / porch / rear-door steps (brick 1:5, provisional) [incl. 20 cft PCC 1:4:8 treads base]	40.0	cft	4.1	23	19	540	-
	Sub-total G			122.6	657	182	15,993	172
H. Septic tank & manholes								
H1	Septic tank excavation (internal 6'-0" x 4'-0" x 5'-6" deep, 2 chambers) [drawing shows ~4'-10" x 3'-5" overall; enlarged to standard size for ~12 users (upper estimate)]	372.9	cft	-	-	-	-	-
H2	P.C.C 1:4:8 3" base	12.0	cft	1.2	6	12	-	-
H3	R.C.C 1:2:4 base slab 6" (#3@6" both ways)	20.6	cft	3.7	9	19	-	29
H4	Brick masonry 1:4 - 9" walls + two 4½" baffle walls	140.0	cft	8.9	45	-	1,890	-
H5	R.C.C 1:2:4 cover slab 5" with 2 manhole openings (#3@6" both ways + trimmers)	13.9	cft	2.5	6	12	-	33
H6	Plaster ¾" 1:3 with water-proofing compound inside septic tank	206.0	sft	4.3	16	-	-	-
H7	Manholes / inspection chambers - 9 nos 2'x2' internal, avg 3'-6" deep: excavation	774.6	cft	-	-	-	-	-
H8	Manholes: P.C.C 1:4:8 4" base	36.7	cft	3.5	18	35	-	-
H9	Manholes: 9" brick walls 1:4	346.5	cft	22.1	111	-	4,678	-
H10	Manholes: benching/channels 1:2:4	13.5	cft	2.4	6	12	-	-
H11	Manholes: inside plaster ½" 1:3	288.0	sft	4.0	15	-	-	-
H12	Manholes: R.C.C cover slabs 3" (#3@6" both ways)	17.0	cft	3.1	8	15	-	54
	Sub-total H			55.7	239	105	6,568	116
I. Overhead water tank (R.C.C, 1000 gallons)								
I1	R.C.C 1:1.5:3 overhead tank internal 8'x5'x4'-6" (1000 imp. gal): 6" base, 6" walls, 4" top [drawings show tank position only (5'-4"x3'-1" symbol) on mumty roof; engineer to confirm support]	106.7	cft	24.4	46	91	-	228
I2	Brick support walls/piers under tank 9" 1:4 (provisional)	45.0	cft	2.9	14	-	608	-
I3	Tank plaster: inside ¾" 1:3 water-proof (197 sft) + outside ½" 1:4 (204 sft)	401.0	sft	6.4	27	-	-	-
	Sub-total I			33.6	87	91	608	228
	GRAND TOTAL (net)			1,480.9	7,020	3,984	118,495	7,679

6. Reinforcement steel schedule (kg, net)

Component	6 mm (#2)	10 mm (#3)	12 mm (#4)	20 mm (#6)	Total (kg)
A. Foundations & Plinth	79	697	502	53	1,330
D. R.C.C work (1:2:4 unless noted)	27	713	1,083	868	2,691
E. Slab reinforcement	-	2,506	580	55	3,141
G. Boundary walls, gates pillars & external works	26	145	-	-	172
H. Septic tank & manholes	-	116	-	-	116
I. Overhead water tank (R.C.C, 1000 gallons)	-	228	-	-	228
TOTAL net	132	4,406	2,165	976	7,679
TOTAL with 5% wastage (to procure)	140	4,630	2,280	1,030	8,080

Bar lengths are generally available in 40 ft (12 m) lengths. Approximate number of bars to buy (with wastage): 10 mm (#3): 679 bars, 12 mm (#4): 189 bars, 20 mm (#6): 38 bars, 6 mm (#2): 47 bars. Per-component detail: footings 673 kg, plinth beams 569 kg, slabs 3,141 kg, concealed beams 1,244 kg, lintels 624 kg, stairs 622 kg.

7. Sewerage, rain-water, water-supply and gas pipes

Horizontal runs were measured on the colour-coded sewerage (sheets 17-20) and sanitary (sheets 21-24) drawings; vertical soil/waste stacks, vent pipes, rain-water pipes, risers and fixture drops were added.

Service	Size	Run / basis	Net (ft)	+10% (ft)	20-ft lengths
Sewerage (UPVC)	5"	Main sewer: rear manholes -> side passage -> septic tank -> road sewer; car-porch MH -> septic tank	156	172	9
Sewerage (UPVC)	4"	WC connections GF/FF/2F (horizontal 47.8 ft) + soil stacks 3x13 ft (FF), 1x25 ft (2F) + 4 vent pipes x 4 ft	128	141	8
Sewerage (UPVC)	3"	Floor traps, basins, showers, sinks (horizontal 116.1 ft) + waste stacks 4x13 ft + 2x25 ft	218	240	12
Rain water (UPVC)	3"	4 rain-water down pipes in 9"x6" ducts (38+29+29+16 ft)	112	124	7
Water supply (PPR)	1"-1¼"	Tank down-take to GF (38 ft) + pump rising main to tank (40 ft) + floor mains (40 ft)	118	130	7
Water supply (PPR)	¾"	Cold water branch lines (measured on sanitary drawings)	370	407	21
Water supply (PPR)	½"	Cold fixture drops 35 points x 4 ft	140	154	8
Hot water (PPR)	¾"	Hot water lines (measured) + geyser risers 36 ft	360	397	20
Hot water (PPR)	½"	Hot fixture drops 20 points x 4 ft	80	88	5
Gas (GI/PPR-gas)	½"-¾"	Gas lines to kitchens, geyser, heaters (measured) + riser from meter 30 ft	190	209	11

7.1 Manholes and septic tank

- 9 manholes / inspection chambers (3 in rear passage, 3 in left side passage, 1 front-left lawn corner, 1 patio, 1 car porch): 2'-0" x 2'-0" internal, average 3'-6" deep, 9" brick walls in 1:4 on 4" P.C.C, benching, ½" plaster 1:3 inside, 3" R.C.C cover slab. Buy 9 manhole covers (C.I. or R.C.C, 20"x20").
- Septic tank (front lawn, connected to main sewer): internal 6'-0" x 4'-0" x 5'-6", two chambers with 4½" baffle walls, 6" R.C.C base, 9" brick walls 1:4, 5" R.C.C top with two 2'x2' openings, ¾" water-proof plaster. Buy 2 covers and 2 x 5" UPVC tees (inlet/outlet).
- Rain water: 4 roof sumps (1'-6" x 1'-6") and 3" pipes in 9" x 6" wall ducts with 1½" tile cover as drawn.

7.2 Fittings (approximate counts)

Group	Fittings
UPVC sewer 5"	Bends 8, Y-tees 6, sockets 9, septic inlet/outlet tees 2
UPVC 4" (soil)	WC connectors 8, bends 20, Y-tees 10, sockets 8, vent cowls 4, clean-out plugs 4
UPVC 3" (waste & rain)	Floor traps (P-traps) 14, gully traps 3, bends 40, Y-tees 20, sockets 15, roof outlets 4, RWP shoes 4
PPR water (cold/hot)	Elbows ~120, tees ~60, sockets ~80, male/female adaptors ~55, unions ~20, ball valves ~20, gate valves 4
Gas	Elbows ~20, tees ~8, gas cocks 8, meter connection set 1
Consumables	UPVC solvent cement 4 kg, PTFE tape, pipe clamps ~120

8. Electrical conduits and boxes

Floor	Light / fan / fancy / profile / CCTV points	Socket points	A.C points	Switch boards
Ground floor	53	41	5	18
First floor	47	30	4	16
Mumty & terrace	15	8	1	5
Total	115	79	10	39

PVC conduit	Basis	Net (ft)	+10% (ft)	10-ft pipes
¾"	Light/fan points 115 x 12 ft + socket points 79 x 8 ft + 25 low-voltage runs x 30 ft (TV, net, CCTV, intercom, bells) + 10% UPS circuits	2,900	3,191	320
1"	DB to 39 switch-boards x 25 ft + 10 A.C points (385 ft) + earthing 30 ft	1,390	1,530	153
1½"	Meter to main DB + risers to FF & 2F DBs	85	94	10

- Back boxes: 39 switch-board boxes (assorted 3"x3" to 12"x3"), ~79 socket boxes, 10 A.C boxes, 13 fan boxes with hooks, ~110 round light junction boxes, 3 DB enclosures (main 12-way GF, 8-way FF, 4-way mumty).
- Accessories: ~300 bends ¾", ~100 bends 1", ~350 couplers ¾", ~150 couplers 1", PVC solvent, GI draw wire.
- Points were counted from the electrical plans (sheets 13-15, legend sheet 16); per-point conduit allowances: 12 ft per light/fan point, 8 ft per socket, 25 ft per switch-board feeder, 30-45 ft per A.C circuit, 30 ft per low-voltage run (TV, internet, CCTV, intercom, bells), +10% for UPS circuits.

9. Optional items (NOT included in the totals)

Item	Description	Cement (bags)	Sand (cft)	Crush (cft)	Bricks / tiles	Steel (kg)	Other
O1	Roof insulation/treatment on terrace & mumty roof (1651 sft): polythene sheet, 4" earth (mud) compacted, 1½" brick tiles in 1" cement mortar 1:4, pointing 1:3	37	170	-	5,720	-	Polythene 1816 sft; earth 660 cft
O2	Underground water tank 1,500 gallons (internal 8'x6'x5'): R.C.C 1:1.5:3 walls/base/top ~170 cft, PCC 12 cft, #3/#4 steel ~400 kg, excavation ~520 cft, inside water-proof plaster	44	94	157	-	400	Excavation ~520 cft
O3	External paving base: side & rear passages + driveway (~430 sft) 3" P.C.C 1:4:8 over 3" sand	10	159	104	-	-	

Roof treatment (item O1) is often counted as part of the grey structure in Faisalabad contracts - add it if your contractor's scope includes it.

10. Points to confirm with your architect / engineer

These observations came up while measuring the drawings. None changes the estimate materially, but each should be settled before work starts.

1. Natural surface level vs road level is marked "as per site". Have it surveyed; it controls excavation depth and earth filling (about 820 cft per 6").
2. Ground-floor M-Bed is 14'-0" x 12'-0" on the structural layout but labelled 14'-3" x 12'-0" on the architectural plan (first floor M-Bed is 14'-3" x 14'-0"). The estimate follows the structural geometry.
3. The septic tank on sheet 17 is only about 4'-10" x 3'-5" overall. For 8 WCs a larger tank (estimated 6'-0" x 4'-0" x 5'-6" internal) is advisable.
4. The overhead water tank is shown only as a symbol (5'-4" x 3'-1") on the mummy roof; its support (walls/beams under it) is not designed. Confirm R.C.C vs plastic tanks.
5. The 6'-6" dining opening has no lintel on the lintel key plan (S-04); an 8'-0" DL-1 has been included.
6. The second stair flight (steps 14-22) passes over the rear car porch (headroom about 7'). The first-floor stair-hall rear wall there is 4½" thick and sits on CB-1 beams cast with the ground-floor roof - make sure these beams and the stair are cast together as detailed.
7. The car-porch front corner (13½" square, hatched on S-01) has no column or footing detail. The estimate assumes an R.C.C column with 4#6 bars on a 3'x3'x1' footing.
8. The ground-floor patio (4'-0" wide) and the first-floor "open 3'-6" wide" shaft are open on the neighbour (property-line) side - confirm whether a privacy/boundary wall is required there (not included).
9. The powder-room wall on the property line is only 4½" thick with a concentric 2'-0" footing (section 4-4) that would cross the property line; consider an eccentric footing like section 1-1.
10. Boundary wall foundations and the back boundary wall are not detailed; confirm the assumed details (Section 4.2).
11. Superstructure mortar ratios are not specified; the estimate assumes 1:5 for 9" and 1:4 for 4½" walls.
12. Sunken slabs (+0'-6" depressed) are noted under first-floor baths - ensure the slab steel is detailed through the drops.

11. Practical checklist for a first project

- Order in stages (Section 2) and keep a stock register; count bricks and weigh steel on every delivery.
- Bricks: first class, uniform size and colour, clear ringing sound, no cracks; soak before laying.
- Cement: check the manufacturing date (use within about a month), store on a raised dry platform, not more than 10 bags high.
- Sand: do a silt (bottle) test - reject if silt is more than about 8%. Chenab sand for concrete, Ravi sand for mortar and plaster.
- Steel: Grade 60 rolling marks on every bar; check weight per foot against the standard (10 mm = 0.376 lb/ft, 12 mm = 0.668 lb/ft, 20 mm = 1.502 lb/ft).
- Measure concrete and mortar with gauge boxes (farma), never by shovels; use a vibrator for slabs and beams.
- Before every slab pour, have the engineer check bar sizes, spacing, cranked bars, extra top bars, covers (¾" slab, 1½" beams) and chairs.
- Place electrical conduits, fan boxes and sewer/water sleeves before casting slabs and before plastering.
- Cure slabs and beams for at least 10-14 days, brickwork and plaster for 7 days.

Appendix A - Measured foundation / ground-floor wall schedule

Coordinates in feet from the rear-left corner of the building (x to the right, y towards the road), measured from structural drawing S-01. Section = foundation type from S-01/S-02.

Wall	Description	Thickness	Section	Length (ft)	Centre-line coordinates (ft)
R1	Back wall (Bed-1 & S-T lobby) facing rear passage	9"	2-2	19.50	(17.38, 0.38) to (36.88, 0.38)
R2	Back wall step (S-T lobby, rear)	9"	2-2	3.31	(17.38, 0.38) to (17.38, 3.69)
R3	S-T lobby / Bed-1 partition	9"	3-3	13.56	(23.13, 0.38) to (23.13, 13.94)
R4	S-T lobby / lobby partition	9"	3-3	5.75	(17.38, 9.25) to (23.13, 9.25)
R5	Bath-Dress & M-Bed / lobby partition	9"	3-3	17.94	(17.38, 7.56) to (17.38, 25.5)
R6	Left outer wall of stair & bath	9"	2-2	9.06	(3.9, 3.69) to (3.9, 12.75)
R6b	Left outer wall jog	9"	2-2	1.27	(2.63, 12.75) to (3.9, 12.75)
R7	M-Bed left outer wall	9"	2-2	12.75	(2.63, 12.75) to (2.63, 25.5)
R8	M-Bed / kitchens partition	9"	3-3	17.25	(1.0, 25.5) to (18.25, 25.5)
R9	Raw kitchen left outer wall	9"	2-2	11.75	(1.38, 25.5) to (1.38, 37.25)
R10	Raw kitchen / TV lounge partition	9"	3-3	8.75	(0.38, 37.25) to (9.12, 37.25)
R11	TV lounge left outer wall	9"	2-2	13.84	(0.38, 37.25) to (0.38, 51.09)
R12	TV lounge front splayed wall	9"	2-2	3.25	(0.38, 51.09) to (3.44, 50.0)
R13	TV lounge front wall	9"	2-2	12.69	(3.44, 50.0) to (16.13, 50.0)
R14	TV lounge / entrance wall	9"	2-2	8.50	(16.13, 41.5) to (16.13, 50.0)
R15	Entrance front wall	9"	2-2	7.00	(16.13, 45.25) to (23.13, 45.25)
R16a	Lobby / dining-drawing partition	9"	3-3	22.94	(23.13, 19.31) to (23.13, 42.25)
R16b	Lobby / car-porch wall	9"	2-2	3.75	(23.13, 42.25) to (23.13, 46.0)
R17	Drawing room front wall (car porch)	9"	2-2	13.75	(23.13, 42.25) to (36.88, 42.25)
R18a	Right property-line wall (rear part)	9"	1-1	19.13	(36.88, 0.38) to (36.88, 19.5)
R18b	Right property-line wall (front part)	9"	1-1	21.19	(36.88, 29.31) to (36.88, 50.5)
R19	Patio top wall	9"	2-2	4.00	(32.88, 19.5) to (36.88, 19.5)

Wall	Description	Thickness	Section	Length (ft)	Centre-line coordinates (ft)
R20	Patio left wall	9"	2-2	4.75	(32.88, 19.5) to (32.88, 24.25)
R21	Patio bottom / P-W top wall	9"	2-2	4.19	(32.88, 24.25) to (37.07, 24.25)
G1	Stair hall rear wall (to back car porch)	4½"	4-4	13.48	(3.9, 3.69) to (17.38, 3.69)
G2	Stair / bath-dress partition	4½"	4-4	13.48	(3.9, 7.56) to (17.38, 7.56)
G3	Bath-dress / M-Bed partition	4½"	4-4	13.48	(3.9, 12.94) to (17.38, 12.94)
G4	Bath / dress partition	4½"	4-4	5.38	(12.32, 7.56) to (12.32, 12.94)
G5	Bed-1 / dress-bath partition	4½"	4-4	13.75	(23.13, 13.94) to (36.88, 13.94)
G6	Lobby / dress partition	4½"	4-4	5.38	(22.94, 13.94) to (22.94, 19.31)
G7	Dress-bath / dining partition	4½"	4-4	9.94	(22.94, 19.31) to (32.88, 19.31)
G8	Dress / bath partition	4½"	4-4	5.38	(27.32, 13.94) to (27.32, 19.31)
G9	Raw kitchen / open kitchen partition	4½"	4-4	11.75	(8.94, 25.5) to (8.94, 37.25)
G10	P-W left wall	4½"	4-4	5.06	(32.69, 24.25) to (32.69, 29.31)
G11	P-W front wall	4½"	4-4	4.19	(32.69, 29.31) to (36.88, 29.31)
G12	P-W right (property-line) wall	4½"	4-4	5.06	(37.07, 24.25) to (37.07, 29.31)
G13	Front elevation projection	4½"	4-4	1.12	(3.5, 50.56) to (4.62, 50.56)
G14	Front elevation projection	4½"	4-4	3.37	(13.13, 50.56) to (16.5, 50.56)
	Total centre-line length			370.71	

Foundation layer	Plan area (sft, union of all walls)	Thickness	Volume (cft)
P.C.C 1:4:8	969.6	3"	242.4
R.C.C footing	833.6	6"	416.8
Brick step 22½"	478.0	6"	239.0
Brick step 18"	523.2	6"	261.6
Brick step 13½"	398.9	12"	398.9
9" plinth strip (plinth beam, DPC, plinth wall)	270.8	9" / 1½" / 2'-9"	203.1 / 33.8 / 744.6

Appendix B - Slab reinforcement bar sets (from S-06, S-07, S-08)

B = bottom bars (alternately straight and cranked, +0.6 ft per bar for hooks/cranks); T = extra top bars (+0.3 ft). No. of bars = distribution width / spacing + 1. Trimmer bars (2#6, 2#4) and chairs are added to each floor total.

Ground-floor roof (S-06) - slab area 1,705.0 sft

Set	Bar	Spacing	Distribution width (ft)	Bar length (ft)	Type	No. of bars	Total length (ft)	Weight (kg)
B1	10 mm	8"	14.59	13.40	Bottom	23	322.0	54.9
B2	10 mm	9"	13.33	14.79	Bottom	19	292.4	49.9
B3	10 mm	7"	14.84	8.25	Bottom	27	238.9	40.8
B4	10 mm	10"	8.58	14.67	Bottom	12	183.2	31.3
B5	10 mm	8"	13.59	16.14	Bottom	22	368.3	62.8
B6	10 mm	6"	16.33	13.54	Bottom	34	480.8	82.0
B7	10 mm	10"	4.58	46.13	Bottom	7	327.1	55.8
B8	10 mm	8"	43.71	5.75	Bottom	67	425.4	72.6
B9	10 mm	6"	46.83	13.75	Bottom	95	1,363.2	232.5
B10	10 mm	10"	12.58	13.57	Bottom	17	240.9	41.1
B11	10 mm	10"	12.58	5.56	Bottom	17	104.7	17.9
B12	10 mm	8"	12.58	14.14	Bottom	20	294.8	50.3
B13	10 mm	10"	12.58	8.61	Bottom	17	156.6	26.7
B14	12 mm	10"	12.96	9.29	Bottom	17	168.1	50.9
T1	10 mm	8"	14.59	4.16	Extra top	23	102.6	17.5
T2	12 mm	10"	15.32	6.50	Extra top	20	136.0	41.2
T3	12 mm	10"	13.44	5.17	Extra top	18	98.5	29.8
T4	10 mm	8"	8.67	6.83	Extra top	15	107.0	18.2
T5	10 mm	8"	4.25	4.50	Extra top	8	38.4	6.5
T6	10 mm	10"	8.58	4.50	Extra top	12	57.6	9.8
T7	10 mm	8"	8.67	13.17	Extra top	15	202.1	34.5

Set	Bar	Spacing	Distribution width (ft)	Bar length (ft)	Type	No. of bars	Total length (ft)	Weight (kg)
T8	12 mm	10"	13.59	7.25	Extra top	18	135.9	41.2
T9	12 mm	10"	3.96	4.58	Extra top	6	29.3	8.9
T10	10 mm	10"	11.58	4.16	Extra top	15	66.9	11.4
T11	10 mm	8"	16.42	12.84	Extra top	26	341.6	58.3
T12	10 mm	10"	12.32	4.91	Extra top	16	83.4	14.2
T13	10 mm	8"	6.96	6.25	Extra top	12	78.6	13.4
T14	10 mm	10"	5.96	3.33	Extra top	9	32.7	5.6
T15	10 mm	10"	17.96	3.92	Extra top	23	97.1	16.6
T16	10 mm	10"	3.15	3.33	Extra top	5	18.1	3.1
T17	10 mm	8"	11.71	6.25	Extra top	19	124.5	21.2
T18	10 mm	10"	22.96	3.92	Extra top	29	122.4	20.9
T19	10 mm	10"	12.58	5.26	Extra top	17	94.5	16.1
T20	10 mm	10"	12.58	8.17	Extra top	17	144.0	24.6
T21	10 mm	10"	12.58	7.17	Extra top	17	127.0	21.7
T22	12 mm	10"	12.96	7.54	Extra top	17	133.3	40.4
					Bar sets total			1,344.3
					Floor total incl. trimmers & chairs			1,435.4

First-floor roof (S-07) - slab area 1,650.7 sft

Set	Bar	Spacing	Distribution width (ft)	Bar length (ft)	Type	No. of bars	Total length (ft)	Weight (kg)
B1	10 mm	8"	14.96	14.31	Bottom	24	357.8	61.0
B2	10 mm	9"	13.58	14.78	Bottom	20	307.6	52.5
B3	10 mm	7"	14.84	8.25	Bottom	27	238.9	40.8

Set	Bar	Spacing	Distribution width (ft)	Bar length (ft)	Type	No. of bars	Total length (ft)	Weight (kg)
B4	10 mm	10"	8.58	14.67	Bottom	12	183.2	31.3
B5	10 mm	8"	13.59	16.15	Bottom	22	368.5	62.8
B6	10 mm	6"	16.33	13.54	Bottom	34	480.8	82.0
B7	10 mm	7"	18.33	11.72	Bottom	33	406.6	69.3
B8	10 mm	8"	18.33	7.10	Bottom	29	223.3	38.1
B9	10 mm	10"	11.58	17.88	Bottom	15	277.2	47.3
B10	10 mm	10"	5.21	19.50	Bottom	8	160.8	27.4
B11	10 mm	8"	14.33	12.00	Bottom	23	289.8	49.4
B12	10 mm	8"	13.21	14.25	Bottom	21	311.8	53.2
B13	10 mm	6"	19.58	14.83	Bottom	41	632.6	107.9
B14	10 mm	8"	13.20	19.00	Bottom	21	411.6	70.2
T1	10 mm	8"	14.96	5.70	Extra top	24	144.0	24.6
T2	10 mm	10"	13.58	4.50	Extra top	18	86.4	14.7
T3	12 mm	10"	13.58	9.17	Extra top	18	170.5	51.6
T4	12 mm	10"	14.96	6.50	Extra top	19	129.2	39.1
T5	12 mm	10"	8.58	5.17	Extra top	12	65.6	19.9
T6	12 mm	10"	10.33	7.83	Extra top	14	113.8	34.5
T7	12 mm	10"	13.59	7.25	Extra top	18	135.9	41.2
T8	12 mm	10"	4.58	4.70	Extra top	7	35.0	10.6
T9	10 mm	10"	11.33	4.16	Extra top	15	66.9	11.4
T10	10 mm	10"	12.32	4.92	Extra top	16	83.5	14.2
T11	12 mm	10"	4.58	7.50	Extra top	7	54.6	16.5
T12	12 mm	10"	12.08	8.84	Extra top	16	146.2	44.3
T13	10 mm	10"	18.33	3.67	Extra top	23	91.3	15.6
T14	10 mm	8"	18.33	6.00	Extra top	29	182.7	31.2

Set	Bar	Spacing	Distribution width (ft)	Bar length (ft)	Type	No. of bars	Total length (ft)	Weight (kg)
T15	10 mm	8"	18.33	6.41	Extra top	29	194.6	33.2
T16	10 mm	10"	17.96	5.67	Extra top	23	137.3	23.4
T17	10 mm	10"	6.21	9.55	Extra top	9	88.7	15.1
T18	10 mm	10"	13.20	4.33	Extra top	17	78.7	13.4
T19	12 mm	10"	18.33	7.16	Extra top	23	171.6	52.0
T20	10 mm	8"	13.44	7.38	Extra top	22	169.0	28.8
T21	10 mm	8"	5.73	4.16	Extra top	10	44.6	7.6
T22	12 mm	8"	14.90	5.00	Extra top	24	127.2	38.5
T23	10 mm	10"	13.20	5.67	Extra top	17	101.5	17.3
					Bar sets total			1,392.1
					Floor total incl. trimmers & chairs			1,446.9

Mumty roof (S-08) - slab area 353.2 sft

Set	Bar	Spacing	Distribution width (ft)	Bar length (ft)	Type	No. of bars	Total length (ft)	Weight (kg)
B1a	10 mm	8"	16.28	7.75	Bottom	26	217.1	37.0
B1b	10 mm	8"	16.67	11.87	Bottom	27	336.7	57.4
B2a	10 mm	10"	10.71	17.25	Bottom	14	249.9	42.6
B2b	10 mm	10"	6.58	16.70	Bottom	9	155.7	26.6
T1	10 mm	10"	16.06	3.33	Extra top	21	76.2	13.0
T2	10 mm	10"	16.06	3.34	Extra top	21	76.4	13.0
T3	10 mm	10"	15.71	3.50	Extra top	20	76.0	13.0
T4	10 mm	10"	15.71	3.50	Extra top	20	76.0	13.0
T5	10 mm	10"	10.71	4.67	Extra top	14	69.6	11.9

Set	Bar	Spacing	Distribution width (ft)	Bar length (ft)	Type	No. of bars	Total length (ft)	Weight (kg)
T6	10 mm	10"	10.71	6.50	Extra top	14	95.2	16.2
T7	10 mm	10"	6.58	4.75	Extra top	9	45.4	7.8
					Bar sets total			251.4
					Floor total incl. trimmers & chairs			259.0

Appendix C - Openings and lintels

Ground floor - openings deducted from brickwork

Opening	Width	Height	Wall	Nos	Wall type
S-T lobby rear door	3'-6"	8'-0"	9"	1	External
S-T lobby to lobby door	3'-6"	8'-0"	9"	1	Internal
Bed-1 door	3'-6"	8'-0"	9"	1	Internal
M-Bed door	3'-6"	8'-0"	9"	1	Internal
Drawing room door (to porch)	3'-6"	8'-0"	9"	1	External
Dining arch opening	6'-6"	8'-0"	9"	1	Internal
Raw-kitchen side door	3'-0"	8'-0"	9"	1	External
Main entrance glazed door (5'-1½" x 20'-6", GF part)	5'-1.5"	11'-0"	9"	1	External
Bed-1 dress door	2'-10.5"	8'-0"	4½"	1	Internal
Bed-1 bath door	2'-6"	8'-0"	4½"	1	Internal
M-Bed dress door	2'-6"	8'-0"	4½"	1	Internal
M-Bed bath door	2'-6"	8'-0"	4½"	1	Internal
Under-stair store door	2'-6"	8'-0"	4½"	1	Internal
Raw to open kitchen door	3'-0"	8'-0"	4½"	1	Internal
Powder room door	2'-3"	8'-0"	4½"	1	Internal

Opening	Width	Height	Wall	Nos	Wall type
Bed-1 window	7'-0"	7'-0"	9"	1	External
M-Bed window	7'-0"	7'-0"	9"	1	External
Drawing window	6'-0"	7'-0"	9"	1	External
Dining window (patio)	3'-3"	7'-0"	9"	1	External
Raw-kitchen window	3'-0"	4'-6"	9"	1	External
TV-lounge front glazing 8'-6" x 20' (GF part)	8'-6"	10'-6"	9"	1	External
Ventilators 2'-6" x 2'	2'-6"	2'-0"	9"	3	External
Exhaust fan openings 1'x1'	1'-0"	1'-0"	9"	3	External

First floor - openings deducted from brickwork

Opening	Width	Height	Wall	Nos	Wall type
S-T lobby door	3'-6"	8'-0"	9"	1	Internal
Bed-1 door	3'-6"	8'-0"	9"	1	Internal
M-Bed door	3'-6"	8'-0"	9"	1	Internal
Front bed door	3'-6"	8'-0"	9"	1	Internal
Store door	3'-0"	8'-0"	9"	1	Internal
Kitchen door	3'-0"	8'-0"	9"	1	Internal
Bed-1 dress door	2'-10.5"	8'-0"	4½"	1	Internal
Bed-1 bath door	2'-6"	8'-0"	4½"	1	Internal
M-Bed dress door	3'-6"	8'-0"	4½"	1	Internal
M-Bed bath door	2'-6"	8'-0"	4½"	1	Internal
Front bed dress door	3'-0"	8'-0"	4½"	1	Internal
Front bed bath door	2'-6"	8'-0"	4½"	1	Internal
Kitchen door/window to terrace 6x8 (sill 6")	6'-0"	7'-6"	4½"	1	External
Bed-1 window	7'-0"	7'-0"	9"	1	External

Opening	Width	Height	Wall	Nos	Wall type
M-Bed window	7'-0"	7'-0"	9"	1	External
Front bed side window	7'-0"	7'-0"	9"	1	External
Front bed glazing (upper part of 20' glazing)	8'-6"	8'-6"	9"	1	External
TV-lounge window (to open shaft)	8'-0"	7'-0"	9"	1	External
Entrance glazing (upper part)	5'-1.5"	8'-6"	9"	1	External
Stair-hall rear windows 3x7	3'-0"	7'-0"	4½"	2	External
Stair-hall vertical glazing slot 2' wide	2'-0"	11'-6.5"	4½"	1	External
Ventilators 2'-6" x 2'	2'-6"	2'-0"	9"	3	External
Exhaust fan openings 1'x1'	1'-0"	1'-0"	9"	4	External

Mumty (2nd floor) - openings deducted from brickwork

Opening	Width	Height	Wall	Nos	Wall type
Stair-hall door to terrace	3'-6"	8'-0"	4½"	1	External
Room door	3'-6"	8'-0"	9"	1	Internal
Bath door	2'-6"	8'-0"	4½"	1	Internal
Laundry door	2'-6"	8'-0"	4½"	1	Internal
Room window 5x7	5'-0"	7'-0"	9"	1	External
Stair-hall rear windows 3x7	3'-0"	7'-0"	4½"	2	External
Stair-hall vertical glazing slot	2'-0"	9'-6.5"	4½"	1	External
Bath ventilator	2'-6"	2'-0"	9"	1	External

Lintel schedule (length = opening + bearing)

Floor	Type	Length	Nos	Section & steel (S-05)
GF	DL-1	5'-0"	5	9"x9", 4#4, #3@6"
GF	DL-1	6'-7.6"	1	9"x9", 4#4, #3@6"

Floor	Type	Length	Nos	Section & steel (S-05)
GF	DL-1	8'-0"	1	9"x9", 4#4, #3@6"
GF	DL-2	4'-4.5"	1	4½"x9", 4#4, #2@6"
GF	DL-2	4'-0"	4	4½"x9", 4#4, #2@6"
GF	DL-2	4'-6"	1	4½"x9", 4#4, #2@6"
GF	DL-2	3'-9"	1	4½"x9", 4#4, #2@6"
GF	WL&DL-1	9'-4.4"	1	9"x9", 4#4, #3@6"
GF	WL-1	8'-8.3"	1	9"x9", 4#4, #3@6"
GF	WL-1	8'-10.1"	1	9"x9", 4#4, #3@6"
GF	WL-1	7'-6"	1	9"x9", 4#4, #3@6"
GF	WL-1	4'-9"	1	9"x9", 4#4, #3@6"
GF	WL-2	10'-4.1"	1	9"x9", 2#4+4#4, #3@6"
GF	VL-1	4'-0"	3	9"x9", 4#3, #3@10"
FF	DL-1	5'-0"	4	9"x9", 4#4, #3@6"
FF	DL-1	4'-6"	2	9"x9", 4#4, #3@6"
FF	DL-1	5'-0"	1	9"x9", 4#4, #3@6"
FF	DL-2	4'-4.5"	1	4½"x9", 4#4, #2@6"
FF	DL-2	4'-0"	3	4½"x9", 4#4, #2@6"
FF	DL-2	5'-0"	1	4½"x9", 4#4, #2@6"
FF	DL-2	4'-6"	1	4½"x9", 4#4, #2@6"
FF	WL-1	8'-10"	3	9"x9", 4#4, #3@6"
FF	WL-1	6'-7.6"	1	9"x9", 4#4, #3@6"
FF	WL-1	8'-8.3"	1	9"x9", 4#4, #3@6"
FF	WL-2	10'-4.1"	1	9"x9", 2#4+4#4, #3@6"
FF	WL-3	3'-7.2"	1	4½"x9", 4#4, #2@6"
FF	WL-3	4'-6"	2	4½"x9", 4#4, #2@6"

Floor	Type	Length	Nos	Section & steel (S-05)
FF	WL-3	8'-1.8"	1	4½"x9", 4#4, #2@6"
FF	VL-1	4'-0"	3	9"x9", 4#3, #3@10"
SF	DL-1	5'-0"	1	9"x9", 4#4, #3@6"
SF	DL-2	5'-0"	1	4½"x9", 4#4, #2@6"
SF	DL-2	4'-0"	2	4½"x9", 4#4, #2@6"
SF	WL-1	6'-9.6"	1	9"x9", 4#4, #3@6"
SF	WL-3	3'-7.2"	1	4½"x9", 4#4, #2@6"
SF	WL-3	4'-6"	2	4½"x9", 4#4, #2@6"
SF	VL-1	4'-0"	1	9"x9", 4#3, #3@10"